

Toward Compact Selected Configuration Interaction Wave Functions with Quantum Monte Carlo—A Case Study of C_2

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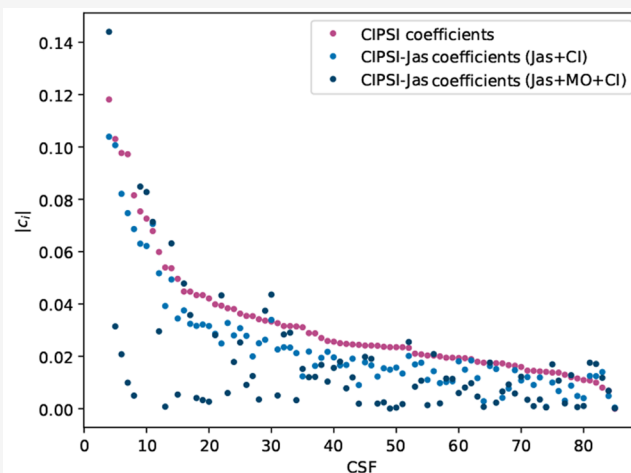


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ABSTRACT: The $^1\Sigma_g^+$ ground state of C_2 is investigated using truncated CIPSI-Jastrow CSF wave functions with Hartree–Fock orbitals within the framework of variational and diffusion quantum Monte Carlo. The truncation is performed based on the absolute value of the CI coefficients, and the Jastrow, molecular orbitals, and CI parameters are either partially or fully reoptimized with respect to the variational energy. Excellent absolute as well as bond dissociation energies are obtained at DMC level with very compact, fully optimized wave functions. By studying the expansions in more detail, we observe a change in the CI picture when reoptimizing the antisymmetric part of the CIPSI-Jastrow wave functions. Furthermore, we demonstrate that a decrease in the VMC energy as well as an improvement of the nodal surface quality can be achieved—with the same expansion size—if the CSFs are selected in the presence of a Jastrow correlation function, laying the foundation for a Jastrow selected CI scheme with quantum Monte Carlo.



INTRODUCTION

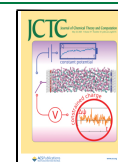
Since the beginnings of modern quantum mechanics in the 1920s, the scope and performance of quantum chemistry have vastly grown. The progress in this field of research has been impressive which is in part due to—but not limited to—the always improving computational architecture. There are a plethora of quantum chemical methods, tailored to the careful description of various problems. A trade-off has, however, always to be made between a satisfactory accuracy and an affordable computational expenditure with the accuracy having to inevitably be forfeited with increasing system size. Although 99% of the total energy of a system can be obtained with the Hartree–Fock method, developed shortly after the formulation of Schrödinger's equation, almost 100 years have, to date, been invested in retrieving the remaining 1%, which is essential for describing chemically relevant properties. Post-Hartree–Fock methods, such as coupled cluster (CC) and perturbation theory as well as density function theory (DFT) are powerful when it comes to characterizing systems with dynamic correlation but they often fail when the static correlation is non-negligible, as e.g. for chemical reactions including bond breaking processes or for open shell systems with partially filled degenerate orbitals. Such multi-reference systems still pose a key challenge for quantum chemistry with intense research efforts striving toward finding a way to, both, accurately describe static and dynamic electron correlation. This problem is traditionally tackled by multi-reference methods, such as

multi-reference configuration interaction (MRCI),^{1,2} complete active space second-order perturbation theory (CASPT2),³ or second-order n -electron valence state perturbation theory (NEVPT2).^{4,5} The last two methods employ multi-configurational zeroth-order wave functions and perform on top of that perturbation theory to correct for the dynamic electron correlation, arising from electronic motion. All of these methods have various drawbacks, ranging from relying on chemical intuition to choose the orbitals that need to be correlated to an exponential scaling with the number of electrons and the basis set size as well as suffering from severe memory requirements.

In order to circumvent the problems presented above, a different class of methods, namely the selected configuration interaction (sCI) approaches—initially pioneered about 50 years ago^{6–8}—has re-experienced an increased focus of research since the beginning of the 21st century due to the shift toward a more parallel programming regime. sCI techniques are iterative methods that probe the full CI

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(FCI) space and add determinants to a reference wave function based on different criteria, such as e.g. energy contributions or determinant amplitudes. Some of the more recently developed algorithms include FCI quantum Monte Carlo (FCIQMC),⁹ adaptive sampling CI (ASCI),¹⁰ and semistochastic heat-bath CI (SHCI).¹¹ The method of choice in this work is CIPSI (configuration interaction using a perturbative selection made iteratively) which was pioneered by Malrieu and co-workers^{7,12} and later revived by Caffarel and co-workers.¹³ The combination of CIPSI wave functions together with the highly accurate quantum Monte Carlo (QMC) methods, more specifically diffusion Monte Carlo (DMC), distinguishing themselves by being vastly parallel and through their favorable scaling with the system size,^{14,15} has proven to be very promising for ground state as well as for excited state energetics.^{16–26} Additionally, the use of a Jastrow correlation function, responsible for the short-range dynamic electron correlation, within QMC has long been established. In related work, a transcorrelated Hamiltonian is employed in FCIQMC.^{27,28}

In this work, we investigate the $^1\Sigma_g^+$ ground state of C_2 with truncated CIPSI-Jastrow wave functions. This molecule is well suited for this kind of analysis because it has only a small number of electrons and displays multi-reference character at the ground state level. For this reason, C_2 is a popular benchmark system that has already been thoroughly investigated in literature, both theoretically and experimentally.^{29–40} We want to highlight two studies that have performed similar investigations to the ones discussed here. First, Clay and Morales³⁵ have investigated C_2 in the equilibrium and a stretched geometry with sCI-Jastrow wave functions with the CSFs being generated using different approaches, not, however, the CIPSI scheme. They also discussed the orbital set choice and found that, both, the CSF selection scheme and the choice of initial orbitals have a severe impact on the fixed-node DMC energy. Second, Giner et al.¹⁸ studied CIPSI-Jastrow wave functions, reoptimized—Jastrow, molecular orbitals (MOs) and determinant coefficients—with VMC, of various first-row atoms. They described the importance of reoptimizing the CI coefficients when a Jastrow factor is used and argued that the MO optimization has only a minor influence on the DMC energies (except for the Be atom). They did not, however, extend their investigation to molecules and, more importantly, to the computation of energy differences. Umrigar et al.⁴¹ showed the importance of reoptimizing the molecular orbital parameters in the presence of a Jastrow factor for Si_2 and C_2 with CASSCF initial wave functions. In this work, we discuss—in contrast to earlier works on sCI-Jastrow-QMC—the importance of going beyond a CIPSI selection since the CI picture is substantially changed in the presence of a Jastrow factor. In addition, we demonstrate the applicability of the (truncated) CIPSI-Jastrow-DMC method for the computation of the bond dissociation energy of C_2 , if one carefully chooses the truncation of the carbon atom. We compare our CIPSI-DMC dissociation energy to very accurate, recently published data. Furthermore, we are able to provide highly accurate C_2 DMC energies with the chosen ansatz and finally, we discuss the impact of the choice of initial orbitals on the expansions.

METHODS

The CIPSI (configuration interaction using a perturbative selection made iteratively) method is a well-established

selected CI approach. The CIPSI wave function corresponds to a linear combination of Slater determinants (SDs) $|D_i\rangle$

$$|\Psi_{\text{CIPSI}}\rangle = \sum_i c_i |D_i\rangle \quad (1)$$

which are iteratively selected—from the full CI space—based on the second-order Epstein–Nesbet perturbative energy correction. For a more detailed description of the CIPSI procedure, we refer to several reviews.^{13,20,42}

Slater–Jastrow-type wave functions are employed for all QMC calculations (except for the comparison with the original CIPSI wave function)

$$|\Psi_T\rangle = e^U \cdot \sum_i c_i |\Phi_i\rangle \quad (2)$$

with the Jastrow correlation function e^U , the coefficients c_i , and the expansions $|\Phi_i\rangle$, which are either configurations state functions (CSFs) or Slater determinants. The Jastrow factor, used in this work, explicitly treats the electron–electron, electron–nucleus, and electron–electron–nucleus correlation.^{43,44} The antisymmetric expansions correspond to products of spin-up and spin-down determinants. Each determinant is constructed from a set of molecular orbitals that are again built from single particle (basis) functions. The trial wave function depends not only on the $3N$ electron coordinates but also on the parameter vector $\mathbf{p} = \{\boldsymbol{\alpha}, \mathbf{c}, \boldsymbol{\kappa}\}$

$$|\Psi_T\rangle \equiv |\Psi_T(\mathbf{R}, \mathbf{p})\rangle \quad (3)$$

which includes the Jastrow parameters $\boldsymbol{\alpha}$, the orbital rotation parameters $\boldsymbol{\kappa}$, as well as the CI coefficients $\mathbf{c}$. The minimization of the VMC energy is achieved by optimizing the different sets of parameters. The optimized trial wave functions are then used as guide functions for the DMC calculations, the results of which allow an assessment of the nodal surface quality of the wave function. In this section, we chose to only introduce the wave function ansatz that we employed throughout this work since elaborate reviews about the VMC and DMC procedures are readily available in literature.^{14,15,45}

COMPUTATIONAL APPROACH

The CIPSI calculations were performed using the Quantum Package⁴⁶ software which works in the basis of Slater determinants. We prefer using CSFs because, first, they have the advantage of being eigenfunctions of the $\hat{S}^2$ operator and, second, the smaller expansions allow for less expensive CI coefficient optimizations with VMC. The spatial symmetries are included as well in the construction of the CSFs. It should be noted, however, that Quantum Package works with spin-adapted wave functions which are obtained by adding all missing determinants—needed to have eigenfunctions of $\hat{S}^2$ —to the internal space. Since Quantum Package does not handle symmetry, which is necessary for the generation of CSFs with spatial symmetry, the HF orbitals were generated with GAMESS⁴⁷ in D_{2h} . The 1s orbitals were kept frozen for the CIPSI calculations. The experimentally derived equilibrium distance of 1.243 Å from Huber and Herzberg⁴⁸ was used. All calculations were performed using the Slater-type triple- ζ all-electron basis set TZPae⁴⁹ by van Lenthe and Baerends. Each Slater-type basis function was expanded into 14 primitive Gaussian-type functions^{50,51} for the GAMESS and Quantum Package⁴⁶ calculations. The CIPSI wave function was truncated at different thresholds, namely 0.0100, 0.0075,

0.0050, and 0.0025 (additional cutoff of 0.0010 for the dissociation energy, see below). The thresholds correspond to the absolute value of the CI coefficients. The QMC calculations were performed within the Slater basis using the electronic structure program Amolqc.⁵² Two different Jastrow factors—denoted sm444 and sm666 in ref 44—with cusp-less three-particle terms were chosen for the calculations.⁴³ The Jastrow factors are denoted Jas-4 and Jas-6, respectively, throughout this work.

The Jastrow, MO, and CI parameters were partially and fully optimized within the VMC framework using the linear⁵³ or the stabilized Newton⁵⁴ method as described in ref 55. For the optimization of the molecular orbitals, the following rotations were considered: doubly to partially occupied, doubly occupied to virtual, partially occupied to virtual, as well as partially to partially occupied. Partially occupied refers to orbitals that are occupied only in some of the CSFs (cf. active orbitals in CASSCF). The orbital rotations are implemented in our code as described in ref 56.

DMC calculations were performed with four different time steps ($\tau = 0.001, 0.002, 0.005, 0.007$) for each wave function and the energies were extrapolated to a zero time step. The extrapolation was performed using a quadratic model.⁵⁷ The Slater determinants were converted to CSFs for each threshold and the calculations were repeated.

The bond dissociation energy of C_2 was calculated for each cutoff at DMC level, using the fully optimized guide functions for C and C_2 , respectively. The dissociation energies were spin-orbit (SO) corrected. The first-order SO correction for the carbon atom was derived from experimental splittings.⁵⁸ C_2 being a closed-shell system, its first-order SO correction is equal to zero.

For the evaluation of the orbital choice, different sets of initial orbitals are compared, namely Kohn–Sham (using the PBE0^{59,60} and the B3LYP,^{61–63} with the VWN(III)⁶⁴ local correlation energy, functionals) and CAS(8,8) orbitals. The computational procedure is identical to the one described for the Hartree–Fock orbitals.

RESULTS

This section is structured as follows: first, the effect of adding a Jastrow factor to the truncated CIPSI wave function is discussed. Then, we contrast SD with CSF expansions, followed by examining the influence of different Jastrow factors on, both, the VMC and DMC energies. Additionally, the DMC absolute as well as dissociation energies of C_2 are compared to literature and the variation of the CI coefficients with regard to the optimization levels is analyzed. We will also briefly review the performance of different sets of initial orbitals. In the last part of this work, we discuss how compact CIPSI-Jastrow wave functions can be obtained, paving the way for a Jastrow-sCI scheme with QMC. All the energies discussed in this work are tabulated in the [Supporting Information](#).

Truncated CIPSI-Jastrow Wave Functions. Post-Hartree–Fock methods aim at incorporating electron correlation in the wave function, which can be achieved in myriad ways. Within—among others—the CIPSI approach, this is traditionally realized through a linear combination of Slater determinants, which includes the static as well as the dynamic electron correlation. Another way, which yields much more compact wave functions, is to capture the dynamic correlation by adding a Jastrow factor to the wave function. With this correlation function, the need for lengthy determinant

expansions vanishes, because in order to capture the short-range dynamic correlation, excitations in higher lying orbitals are necessary. This is demonstrated by [Figure 1](#), where the truncated CIPSI wave functions are compared to Jastrow optimized ones.

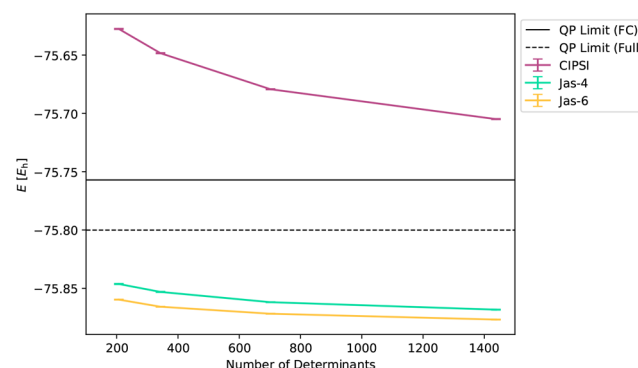


Figure 1. VMC energies of truncated CIPSI wave functions with and without Jastrow factor as a function of the number of determinants. The QP limit (FC) corresponds to the estimated frozen core full CI limit $E + E_{PT2}$ in the given basis. The QP limit (Full) includes the correlation of the 1s orbitals; see text. The data points are connected as guidance for the eyes.

The energies of the truncated CIPSI wave functions (pink data) converge monotonically toward the full CI estimate (solid line in [Figure 1](#)) for the given basis set. By including a Jastrow correlation function, the energy is substantially lowered by about $0.2 E_h$ for each threshold, respectively. The CIPSI wave function contains 1,985,516 determinants. The QP limit (FC) energy originates from a frozen core CIPSI calculation. Since the QMC calculations with a Jastrow factor correlate all electrons, the estimated CIPSI full CI limit, correlating all electrons, is calculated with Quantum Package, as well (dashed line in [Figure 1](#)). Note that the CIPSI-Jastrow VMC energies are substantially lower than this limit, already for 200 determinants. The higher-order Jastrow factor provides lower energies due to more ample flexibility in the wave function.

The accuracy of VMC being limited by the wave function form and flexibility,¹⁵ we intend to investigate how the CIPSI-Jastrow wave functions perform within the more accurate DMC framework. This method allows us—after a zero time step extrapolation—to assess the quality of the nodal hypersurface of a given wave function, the accuracy being determined by the fixed-node error. This will be discussed later in this study. In addition, the robustness of a given ansatz stands and falls with how accurately it can determine molecular properties, such as the dissociation energy. Since the truncated CIPSI-Jastrow approach most likely suffers from size inconsistency, which cannot be neglected when computing energy differences, we will also discuss different truncation schemes that allow for accurate dissociation energies of C_2 .

Comparison of Determinant and CSF Expansions. Although Quantum Package⁴⁶ employs spin-adapted wave functions, we opted for the use of configuration state functions in this work for two different reasons. First, by cutting off the CIPSI wave function, it can no longer be ensured that the truncated expansion is an eigenfunction of the $\hat{S}^2$ operator and second, by constructing CSFs from determinants, the length of

the expansion can be substantially reduced (see Table 1), which leads to computationally less demanding optimizations

Table 1. Number of Determinants and CSFs for the Different Thresholds

threshold	no. determinants	no. CSFs	no. determinants (re-expanded)
0.0100	207	85	337
0.0075	343	138	575
0.0050	702	266	1090
0.0025	1438	528	2242

with VMC. For the CSFs, the thresholds in Table 1 correspond to the determinant cutoffs. The CSFs are generated for all the determinants that are above the respective thresholds. As for the missing determinants, needed to complete the CSFs: their coefficients are obtained from the original CIPSI wave function. The last column in Table 1 gives the number of determinants if the CSFs are re-expanded into the SD basis.

In Figure 2, we compare the VMC energies of Slater determinant and CSF expansions for different sets of optimized

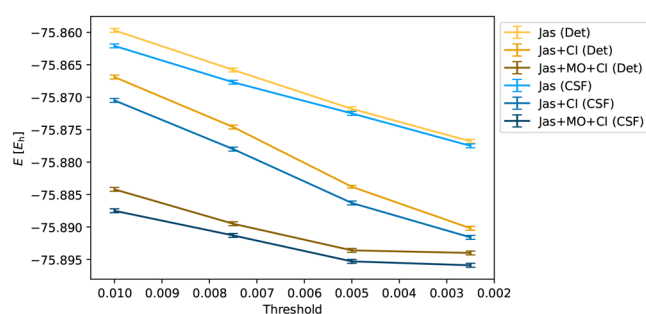


Figure 2. VMC energies of determinant and CSF CIPSI-Jastrow wave functions at various optimization levels computed for different thresholds. An sm666 Jastrow factor was used. The data points are connected as guidance for the eyes.

parameters. First of all, we see that with every additional set of parameters that is optimized, a significant decrease in energy can be observed for each threshold, regardless of the choice of expansion. Furthermore, the VMC energies decrease monotonically as a function of the expansion length, which is inherent since the wave functions have higher flexibility due to more variational parameters being included. The CIPSI-Jastrow wave functions with CSFs yield lower VMC energies compared to the ones using SDs for all thresholds, irrespective of the optimization level. After expanding the CSFs back to determinants—see last column in Table 1—the wave functions contain more determinants than the original ones. Shorter CSF expansions leading to lower energies, hence, usually implies computationally more demanding calculations. The optimization of the CI coefficients—usually a very time-consuming part of the VMC calculations—remains, nonetheless, cheaper for CSF wave functions. The Jastrow correlation function having a nonzero overlap with the determinant expansion, it is essential to reoptimize the expansion coefficients (as well as the MO parameters). This is corroborated by a previous study of ours which focused on the computation of accurate dissociation energies for transition metal dimers⁶⁵ as well as by many other works.^{18,31,33,66} It should be noted, that the Jas+CI curves are much steeper than the ones for the fully optimized wave functions, which seem to be close to convergence already.

Since the DMC energies will be discussed in more detail below within the framework of different Jastrow factors, we dispense, here, with an exhaustive discussion and refer to the Supporting Information for a more elaborate evaluation. In summary, the CSF-based wave functions are slightly favored compared to the SD-based ones in terms of nodal surface quality.

Influence of Different Jastrow Factors. Having, in the previous section, established the importance of using a Jastrow correlation function, that explicitly handles electron correlation, we aim, here, to investigate whether and how the size of the Jastrow factor impacts the wave function optimization with VMC as well as the quality of the nodal surface, studied with DMC. Figure 3 introduces the VMC energies for two Jastrow factors and different optimization levels.

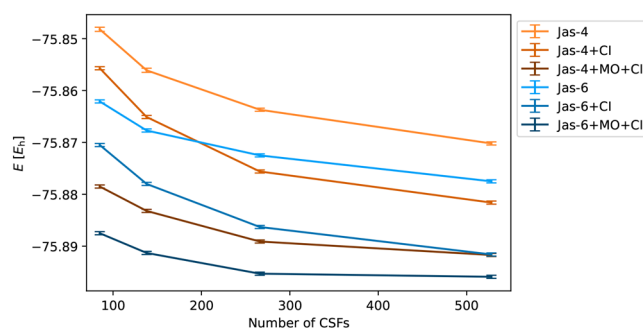


Figure 3. VMC energies of truncated CIPSI-Jastrow wave functions at various optimization levels as a function of the number of CSFs. The order of the Jastrow factor is given by Jas-X. The data points are connected as guidance for the eyes.

For the fully optimized wave functions, the energies appear converged, while for the Jas and Jas+CI optimization levels, we anticipate that lower energies are obtained for larger expansions. Furthermore, the larger Jastrow factor (Jas-6) provides lower VMC energies for each optimization level, which is expected, since a higher amount of variational parameters goes hand in hand with a more flexible wave function. The decrease in energy is largest when moving from the Jas to the Jas+CI optimization level, which also yields the steepest energy curves. Giner et al.¹⁸ investigated CIPSI-Jastrow wave functions for several atoms using HF initial orbitals and they found no significant improvement of their VMC energies by also optimizing the molecular orbitals. For C₂, this is clearly not the case. By also including the orbital optimization, the VMC energies are lowered by 5 to more than 15 mEh, depending on the cutoff value.

In Figure 4, the DMC energies of the optimized CIPSI-Jastrow wave functions are compared to the DMC energies obtained with the truncated CIPSI wave functions. In the DMC—in contrast to the VMC—framework, one does not necessarily observe a systematic lowering of the energy with an increasing expansion size. This has been reported by various studies.^{67–69} DMC provides the exact ground state energy of a system with the nodes of the trial wave function as a boundary condition. There is no guarantee that this energy is approached monotonically with growing expansion size. Figure 4 demonstrates, however, that the DMC energies reported here, and throughout this work, decrease monotonically within the statistical uncertainty with increasing expansion size—this is also the case for the CIPSI nodes—if the CI (and MO)

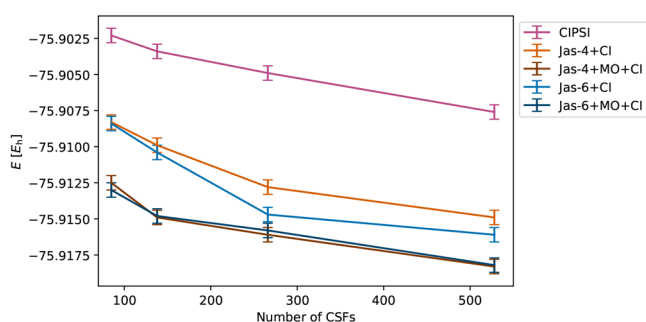


Figure 4. DMC energies of truncated CIPSI wave functions at various optimization levels as a function of the number of CSFs. The order of the Jastrow factor—if used—is given by Jas-X. The data points are connected as guidance for the eyes.

parameters are reoptimized in the presence of the Jastrow factor. This has already been observed in a study by Giner et al.¹⁸ Additionally, for each threshold, the CIPSI nodes yield significantly higher energies than the nodes obtained by reoptimizing the CIPSI-Jastrow wave functions. To ensure, however, a fair comparison between the CIPSI method—unfolding its true potential only when using large expansions—and the CIPSI-Jastrow procedure, the two approaches are compared in the next section. As for the different optimization levels, the larger Jas-6 factor is superior for the Jas+CI optimization for the smaller cutoff values. By also including the MO optimization, both Jastrow factors provide similar energies for each threshold.

Although the VMC energies of the fully optimized wave functions appeared converged for the larger expansions (see Figure 3), the DMC results show that the nodal surfaces are clearly not converged. Note that, although the VMC energies of the Jas-4+MO+CI and the Jas-6+CI wave functions with the smallest cutoff coincide, see Figure 3, the nodal surface of the former wave function is much better. Thus, the performance of a given approach within VMC cannot necessarily be translated to DMC.

Comparison to Literature. In this part of the study, we intend to weigh various methods against one another in order to evaluate the potential of the CIPSI-Jastrow ansatz. In Figure 5, the absolute energies of C_2 are compared for different methods. The J-VBSCF DMC energy is taken from ref 33. The

DMC/CAS(8,8) and the MRCI(Q) energies originate from the works of Barnett et al.⁷⁰ and Peterson et al.,⁷¹ respectively. The DMC/CAS(8,8) (Jas+MO+CI) energy was computed in this work. The DMC/CIPSI value is obtained from ref 19, with the CIPSI wave function comprising 173,553 Slater determinants. Toulouse and Umrigar⁵⁵ have extrapolated their C_2 DMC energies—obtained for several truncated wave functions—to the limit that all CSFs (of the considered CI space) are included in the wave function. The extrapolated value corresponds to the DMC/RAS one in Figure 5. This procedure, adopted in several other works^{35,41} as well, makes use of the fact that the original wave function is normalized to 1. By plotting the DMC energies of the fully optimized truncated CIPSI-Jastrow wave functions with respect to the sum of the squares of the CI coefficients, one can extrapolate the DMC energy to 1 using a linear fit. We used five truncated CIPSI-Jastrow wave functions for the extrapolation; the graph of the extrapolation can be found in the Supporting Information. The additional DMC energy (computed for the discussion of the bond dissociation energy, see below) of the fully optimized wave function using a cutoff value of 0.0010 (1766 CSFs) corresponds to $-75.9197(5) E_h$. The energy is not displayed in Figure 5 for clarity reasons. The extrapolated DMC energy for the CIPSI-Jastrow approach is also given in Figure 5. The *estimated exact* value originates from Bytautas and Ruedenberg³⁰ and is derived from Table 10 in ref 30 by adding the *core correlations* amount to the *nonrelativistic valence-correlated* energy.

When studying Figure 5, several observations can be made. First, the DMC/CIPSI (no Jas) energy from Caffarel and co-workers¹⁹ is about 10 to 15 m E_h lower than the ones computed for the truncated CIPSI wave functions (pink data, no Jas), which emphasizes that by truncating CIPSI expansions and dispensing with a Jastrow factor, one loses a substantial amount of the correlation energy. Remarkably, the DMC energy of the fully optimized CIPSI-Jastrow wave function with the largest threshold (85 CSFs, dark blue) is lower than that of the DMC/CAS(8,8) (Jas+MO+CI), which contains 168 CSFs. This corroborates the fact that excitations beyond the full valence CAS are important, which has been shown in other studies, as well.^{22,41} In addition, the energy of the largest CIPSI-Jastrow wave function at the Jas+CI optimization level (528 CSFs, blue) already equals the DMC/CIPSI energy of

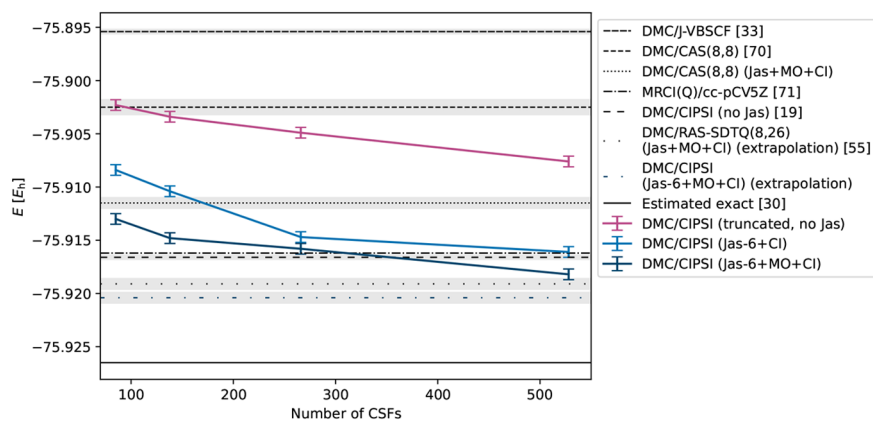


Figure 5. Comparison of CIPSI-Jastrow-DMC energies, computed in this work (pink and blue data), with different methods and ansätze from literature. The data without indicated references were computed in this work. For references, see text. Error bars are indicated in gray. The data points are connected as guidance for the eyes.

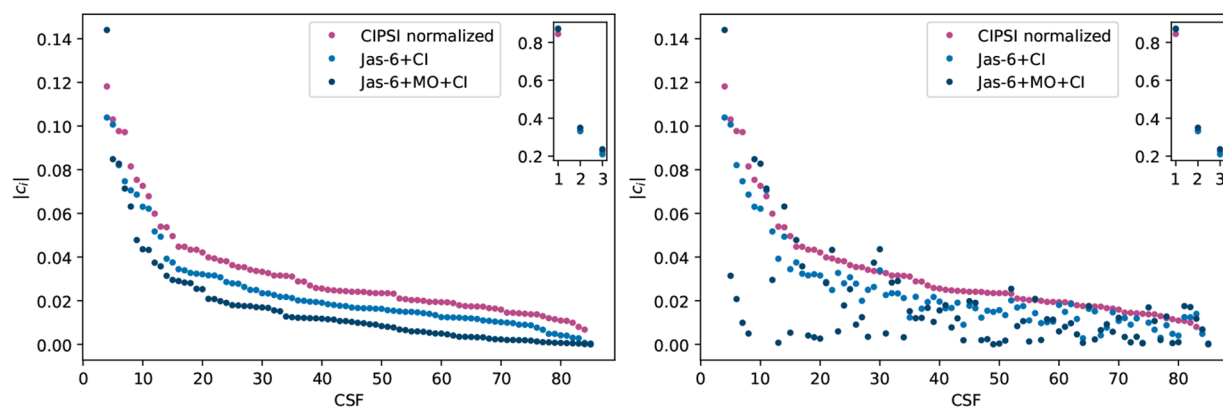


Figure 6. Absolute values of the CSF coefficients for CIPSI wave functions with and without Jastrow factor truncated at a threshold of 0.0100. The first three coefficients are depicted in the subplots. Left: all coefficients are sorted from largest to smallest. Right: the original CIPSI coefficients (pink) are sorted from largest to smallest. For the CIPSI-Jastrow wave functions, the coefficients are sorted such that the CSFs are identical for each index, i.e. x -coordinate. The CSFs are normalized for each approach.

Caffarel and co-workers with 173,553 determinants. Furthermore, the fully optimized CIPSI-Jastrow wave function with the largest expansion (528 CSFs, dark blue) provides a DMC energy that is considerably lower than both the MRCI(Q) and the DMC/CIPSI data and that is close to the extrapolated DMC/RAS energy from Toulouse and Umrigar. The DMC energy obtained for the fully optimized wave function with a threshold of 0.0010 (see above) is slightly lower than the extrapolated DMC/RAS one, which corresponds to $-75.9191(5) E_h$. The extrapolated DMC/CIPSI-Jas-6+MO+CI energy amounts to $-75.9204(5) E_h$ which is about 1 m E_h lower than the one from the DMC/RAS ansatz. These findings illustrate that the CIPSI-Jastrow-DMC approach is promising since it allows for accurate energies using very compact wave functions.

CI Coefficients. Having seen in the previous parts of this work that reoptimizing the CI coefficients in the presence of a Jastrow correlation function has a considerable impact on, both, the VMC and DMC energies, it is worth taking a look at how the coefficients are altered by the optimization. The investigation is done exemplarily for the smallest wave function. After the CI optimization, the sum of the squares of the CI coefficients are renormalized to 1 (ignoring the Jastrow factor). The results are presented in Figure 6.

The absolute values of the CI coefficients are diminished—starting from the fifth one—if a Jastrow factor is included in the wave function, see left graph. This nicely corroborates the effect of the Jastrow factor, namely that it—rather than the expansion—accounts for the short-range dynamic electron correlation. The CI expansion being normalized, the CI coefficients—and thus the weights—of the first few CSFs are necessarily increased by adding a Jastrow factor. The decrease of the absolute value of the CI coefficients is even more distinct for the fully optimized wave function which shows that the MO optimization has a major impact on the CI coefficients and hence on the nodal surface. The right graph in Figure 6 illustrates how the coefficients are altered by the optimization for each CSF. For the Jas+CI optimization, the coefficients are much closer in value to the original CIPSI coefficients than for the full optimization. Additionally, the trend for the Jas-6+CI coefficient decrease is similar—with minor fluctuations—to the one for the normalized CIPSI coefficients. Considering, now, also the optimization of the molecular orbitals, the

picture changes drastically, especially for the lower index CSFs, which have—with a few exceptions—significantly smaller coefficients after the full wave function optimization. On the other hand, the coefficients of several CSFs are increased by the Jas+MO+CI optimization. This shows that the selection of the CSFs is influenced by the optimization which emphasizes that choosing configurations in the presence of a Jastrow correlation function is important. In order to investigate this in more depth, we decided to closer analyze the CSFs, whose coefficients are drastically altered by the full optimization of the wave function, by looking at which excitations are favored or, contrarily, less likely. We focused on the first 30 CSFs (of the fully optimized wave function), the coefficient changes being the most distinct for those configurations. Concerning the CSFs that gain in importance by the optimization: double (favored) and quadruple excitations occur predominantly, with $\pi \rightarrow \pi^*$ excitations being the most important. For the CSFs with smaller coefficients for the optimized wave function: $\sigma \rightarrow \sigma$ as well as several mixed double excitations are disfavored. We observed a significant contribution from the d_{xz} and d_{yz} orbitals of both carbon atoms to different π^* orbitals, see Supporting Information, and judge this to be the reason why excitations in those orbitals are less likely.

Bond Dissociation Energy. We have shown, so far, that the CIPSI-Jastrow ansatz is able to provide accurate absolute energies for C_2 . As chemists, we are, however, primarily interested in computing molecular properties of a given system, which is why we now focus on how well the CIPSI-Jastrow-DMC method performs for the electronic dissociation energy D_e of C_2 . The CIPSI method, usually including several hundred thousand up to more than a million determinants, depending on the system at hand, is quasi size consistent. A difficulty that arises for the truncated CIPSI-Jastrow approach is that it probably suffers from size inconsistency since the chosen cutoffs are rather aggressive. In order to assess this, we analyzed different truncation schemes for the carbon atom wave function. The results are listed in Table 2. The VMC and DMC energies for the carbon atom can be found in the Supporting Information. The dissociation energies are first-order spin-orbit corrected, the correction (for the carbon atom) being determined from experimental splittings and amounting to -0.003669 eV.⁵⁸ First, it should be noted that the best estimate computed in this work is 6.39(2) eV. This

Table 2. Computed DMC Bond Dissociation Energies of C_2 in eV at the Jas-6+MO+CI Optimization Level with Different Truncation Schemes for the Carbon Atom Wave Functions^a

threshold C_2	threshold C^b	no. CSFs	D_e	threshold C^c	no. CSFs	D_e
0.0100	0.0100	40	6.16(2)	0.1000	2	6.52(2)
0.0075	0.0075	46	6.18(2)	0.0550	5	6.68(2)
0.0050	0.0050	55	6.20(2)	0.0380	13	6.37(2)
0.0025	0.0025	94	6.21(2)	0.0340	19	6.40(2)
0.0010	0.0010	186	6.35(2)	0.0230	26	6.39(2)

^aThe number of CSFs for the different wave functions is given, as well. The dissociation energies are first-order spin-orbit corrected; see text for details. ^bSame threshold for C and C_2 . ^cSums of squared CI coefficients of the C wave functions were chosen such that they are as close as possible to the sums of squared CI coefficients of the C_2 wave functions.

value should thus be used as a reference for the discussion of the truncation schemes. We decided to add an additional (smaller) threshold for the computation of the dissociation energy in order to assess whether a convergence can be observed. For the first scheme, we simply used the same coefficient threshold for the C atom wave function as for the dimer. Apart from the smallest cutoff, the dissociation energies (first D_e column) are not very accurate. This is not surprising, since by choosing equal thresholds, we achieve a much better description of the atom than of the molecule, which is mirrored by the dissociation energy being severely underestimated. In the second ansatz, we determined the sum of the CSF weights (squares of the CI coefficients) after the truncation. The carbon atom wave functions were truncated such that the normalization was as close as possible to the one of C_2 . For this scheme, we observe a nice convergence of the dissociation energy. This approach is superior since we expect that a similar amount of correlation is recovered for, both, the atom and the molecule. It should be noted that for the smallest C_2 cutoff, the dissociation energies for both truncation schemes are in good agreement with each other. In order to assess whether this is based on the expansions being large enough to, both, accurately describe the atom and the molecule or whether this is a mere coincidence, the dissociation energies should be evaluated for an even smaller C_2 threshold and the respective carbon thresholds. In a next step, we will compare the best estimate of the dissociation energy from this study to data from literature (see Table 3). The zero-point energy, necessary for D_0 , is taken from ref 72 and amounts to 0.1146 eV.

There are myriad values in literature for the bond dissociation energy of C_2 . In this work, we focus on more recent and highly accurate experimental as well as theoretical data. The dissociation energy from Borsovsky et al.⁴⁰ is, to the best of our knowledge, the best experimental estimate currently available. Table 3 shows that our dissociation energy fits well into the range of recent experimental estimates and that it is in agreement with (most) theoretical values. The *i*-FCIQMC approach underestimates the experimental dissociation energy. The procedure for the computation of the DMC/CIPSI-Jastrow (this work) and the DMC/CAS(8,8) values is the same, apart from the choice of expansion. By employing a CIPSI initial expansion—in contrast to a CASSCF wave function—we were able to improve the bond dissociation energy of C_2 . What is more, the dissociation energy computed

Table 3. Calculated or Measured Bond Dissociation Energies of C_2 in eV, with and without Zero-Point Correction

investigators	method	D_e	D_0
this work	DMC/CIPSI(Jas-6+MO+CI)/TZPae	6.39(2)	6.28(2)
Karton ³⁸	CCSDTQ567/CBS		6.248
Haghighi Mood ⁷³	DMC/CAS(8,8)(Jas+MO+CI)/TZPae	6.433(16)	6.315(16)
Umrigar et al. ⁴¹	DMC/CAS(8,18)(Jas+MO+CI)/BFD-VSZ	6.36(1)	
Toulouse and Umrigar ³¹	DMC/CAS(8,8)/CVB1 full optimization ^a	6.482(3)	
Cleland et al. ⁷⁴	<i>i</i> -FCIQMC/VQZ+ $\Delta E_{F12}^{csd(T)}$	6.296(9)	
Hermann et al. ³⁷	CASPT2-F12/cc-pVTZ-F12	6.370	6.257
Ruscic ⁷⁵	exp (ATcT)		6.245(3)
Borsovsky et al. ⁴⁰	exp (VMI and MPI)		6.2476(3)
Urdahl et al. ²⁹	exp (LIF)		6.30(2)
Visser et al. ³⁹	exp (TC-RFWM)/ icMRCI+Q/cc-pCVQZ-F12	6.36(2)	6.25(2)
Bytautas and Ruedenberg ³⁰	exp (from ATcT, atomic ionization energies)	6.44(2) ^b	6.30(2)

^aOptimization of the Jastrow, CSF, orbital, and basis exponent parameters. ^bNonrelativistic estimate. Corresponds to sum of nonrelativistic valence-correlated and core correlations values; see Table 10 of ref 30. Relativistic estimate ($D_0 + ZPE$) corresponds to 6.42(2) eV.

in this work agrees well with the CASPT2 and the DMC/CAS(8,18) data. Toulouse and Umrigar³¹ report a DMC dissociation energy of 6.482(3) eV, obtained using a full valence CAS approach with a fully optimized wave function, including the optimization of the basis function exponents. This bond dissociation energy overestimates the theoretical and experimental ones. Finally, the coupled cluster value from Karton, keeping in mind the computational expenditure needed for these calculations, shows an excellent agreement with the experiment.

Choice of Initial Orbitals. So far, all of the results that have been presented were obtained using Hartree–Fock orbitals. Various studies have shown that the use of Kohn–Sham orbitals is favorable in combination with quantum Monte Carlo.^{55,76–80} Several studies,^{23,24,26} making use of the CIPSI-Jastrow approach, employ CASSCF or even natural orbitals to construct the CIPSI expansions. In this part of our study, we compare the performance of different sets of initial orbitals with respect to the DMC energies of C_2 (see Figure 7). The QMC calculations for this part of the work were conducted in the Slater determinant basis. Since only trends are discussed, we judge the results to be alike for the CSF basis. The VMC energies of the Jas+CI and the fully optimized CIPSI-Jastrow wave functions using different initial orbitals are shown in the Supporting Information. The HF orbitals provide the highest VMC energies in every case. At the Jas+CI level, the B3LYP KS orbitals are slightly favored while for the full optimization, the VMC energies are similar (except for HF), irrespective of the initial orbital choice.

In 2019, Per and co-workers argued that an accurate density is important in order to obtain good nodal surfaces.⁸¹ They found that the PBE0 (followed by the B3LYP) functional provided the best results for the systems that they investigated. Comparing the VMC and DMC energies of C_2 using initial

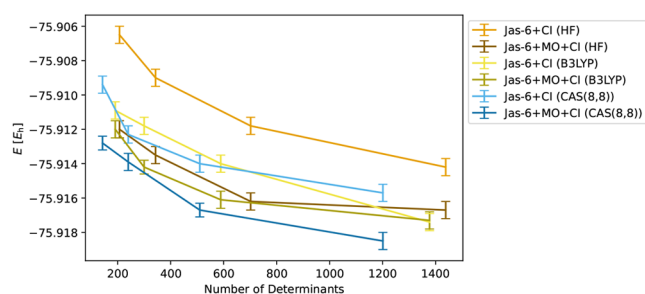


Figure 7. DMC energies of truncated CIPSI-Jastrow wave functions as a function of the number of determinants for different sets of initial orbitals. For each plot, the data points correspond to the $|c_i|$ cutoff values of 0.0100, 0.0075, 0.0050, and 0.0025. An sm666 Jastrow factor was used. The data points are connected as guidance for the eyes.

PBE0 and B3LYP Kohn–Sham orbitals, respectively, we found that B3LYP provided superior results. The PBE0 DMC energies are not shown in Figure 7 for clarity reasons, the comparison to B3LYP can however be found in the Supporting Information. Several observations can be made regarding Figure 7. First, for a given cutoff value, the size of the expansion is reduced—compared to HF—when using KS orbitals. This is even more prominent when CASSCF orbitals are employed. By plotting the absolute value of the CI coefficients as a function of the determinant indices, see Supporting Information, one observes a steeper descent of the CIPSI coefficients when using CASSCF orbitals than for the HF and B3LYP counterparts. This indicates that some correlation is—compared to HF—already included in the orbitals, the expansion, therefore, being shorter for a given threshold. For wave functions constructed from the same determinants, assuming that the optimization works well, the choice of initial orbitals would make no difference to the final VMC and DMC energies. The differing number of determinants for a given cutoff in Figure 7 thus implies that different determinants are chosen, depending on the choice of initial orbitals. Second, at the Jas+CI optimization level, the expansions using B3LYP KS orbitals yield overall lower DMC energies than the wave functions computed with the CASSCF counterparts. Remarkably, for the largest expansion using KS orbitals, the energy is not improved by optimizing the MO parameters. For larger systems, therefore, it seems promising to omit the costly MO optimization when using KS orbitals. The fully optimized CIPSI-Jastrow approach with CASSCF orbitals provides the lowest energies for each threshold. This procedure has, however, the disadvantage of needing to choose an active space for the generation of the orbitals, which stands in the way of the CIPSI-Jastrow-DMC method becoming a black box approach. It should be noted that, ideally, one would employ orbitals that are optimized within the selected CI framework, which has been studied by Yao and Umrigar.⁸²

Clay and Morales³⁵ investigated C_2 in both the equilibrium and a stretched geometry with QMC using multi-Slater-Jastrow wave functions. They argued that using orbital sets that already take into account electronic correlation for the generation of the configurations is important, especially because one observes a faster convergence with the number of included CSFs. This significance of the initial orbital choice is nicely corroborated by our findings. Cuzzocrea et al.²⁵ recently published excitation energies for cyanine dyes using CIPSI-Jastrow wave functions. They found that the use of

different initial orbitals has no significant impact on the excitation energies. The arguments we have made above are only valid for the absolute energies of C_2 . It remains to be seen whether the choice of initial orbitals has an impact on the dissociation energy of C_2 , which we are currently investigating.

Jastrow-sCI Scheme. One goal of this work is to provide an approach that allows for the generation of compact selected configuration interaction wave functions. Having established the validity of the CIPSI-Jastrow-DMC procedure in the previous parts of this study, we now intend to show the importance of moving beyond the original CIPSI selection. Since the CI picture is substantially altered when simultaneously optimizing all the wave function parameters (see discussion of the CI coefficients), we judge the selection of the CSFs in the presence of a Jastrow factor to be of great importance. Before discussing this in more detail, however, let us first compare the full valence CAS and the CIPSI expansions (see Table 4). The CAS(8,8) wave function contains 168

Table 4. VMC and DMC Energies in E_h for Different Wave Function Ansätze^a

	CAS(8,8)	CIPSI
E	−75.6381 ^b	−75.6751(4) ^c
$E_{\text{VMC}}(\text{Jas-6})$	−75.8594(3)	−75.8707(3)
$E_{\text{VMC}}(\text{Jas-6+CI})$	−75.8696(3)	−75.8820(2)
$E_{\text{VMC}}(\text{Jas-6+MO+CI})$	−75.8792(3)	−75.8937(3)
$E_{\text{DMC}}(\text{Jas-6+MO+CI})$	−75.9115(5)	−75.9151(5)

^aThe CAS(8,8) wave function contains 168 CSFs. The CIPSI expansion was truncated at 168 CSFs. ^bCASSCF energy. ^cCalculated with VMC.

CSFs, so, in order to compare both ansätze, the CIPSI expansion is truncated, after sorting with respect to $|c_i|$, at 168 CSFs. Table 4 demonstrates that, irrespective of the optimization level, CIPSI is superior—at least for C_2 —to CAS in terms of energy. Additionally, the fully optimized CIPSI-Jastrow wave function has a better nodal surface, demonstrated by the considerably lower DMC energy. These findings are not surprising, since CAS is known to include various CSFs that do not significantly contribute to the correlation energy. CIPSI, on the other hand, chooses expansions based on an energy contribution. This was already shown by Dash et al. in their work on 1,3-*trans*-butadiene.⁸³ The comparison between CIPSI and CAS justifies the transition from CAS initial wave functions—used in several of our previous studies^{65,79}—to CIPSI expansions, leaving behind the limitations of CASSCF and paving the way toward the investigation of larger systems.

Let us now move on to the discussion of the sCI scheme in the presence of a Jastrow correlation function (Jas-sCI). In this part, we aim at discussing the selection of the CSFs—more precisely, we want to compare different selection schemes by discussing the effects of the Jastrow factor and of the molecular orbitals. All the energies, portrayed in Tables 5, 6, and 7, are obtained for wave functions with 168 CSFs. While selected CI usually refers to iterative selection schemes, we present, in this study, only one CI selection step which may, in subsequent work, be the basis for a full iterative Jastrow-sCI algorithm.

First of all, we want to contrast (see Table 5) the effect of adding a Jastrow factor and the choice of molecular orbitals. Table 5 reveals that the nodal surface is improved—for the same expansion size—by using orbitals from a correlated (here

Table 5. DMC Energies in E_h for Different Wave Function Ansätze^a

	E_{DMC}
CIPSI/HF	−75.9047(5)
CIPSI/CAS	−75.9061(5)
CIPSI-Jas/HF (Jas-6+CI)	−75.9121(5)

^aThe expansions were truncated at 168 CSFs. The CIPSI entries correspond to pure CIPSI wave functions. For the CIPSI-Jas wave function, the Jastrow and CI parameters were optimized. The initial orbitals—used for the CIPSI calculation—are either Hartree–Fock or CASSCF orbitals.

CASSCF) calculation as opposed to using HF orbitals in the CIPSI calculation. The effect is, however, much smaller compared to the approach of adding a Jastrow correlation function to the CIPSI HF wave function and optimizing the Jastrow and CI parameters. It should be mentioned that the CIPSI/HF and CIPSI/CAS DMC energies were calculated with an optimized Jastrow factor in order to increase the efficiency of the calculations. Note, however, that the Jastrow factor does not change the nodes of the wave functions and thus the DMC energies.

In a next step, we selected the CSFs in the presence of the Jastrow factor, see Table 6. For the Jas-sCI entries, we started with larger expansions (initial cutoff) at the Jas-6+CI initial opt level, truncated the wave functions (after the optimization and sorting) to 168 CSFs, and subsequently reoptimized the Jastrow and CI parameters. The HF orbitals remain unaltered for this Jas-sCI procedure in order to limit the evaluation to the impact of the Jastrow factor. For the smaller initial wave function (0.0050), a lower VMC energy is obtained in the Jas-sCI scheme while the DMC energy is similar to the DMC CIPSI-Jas one. As for the larger initial expansion, a decrease in energy is observed for, both, the VMC and DMC energies compared to the CIPSI selection. This shows that one can get lower energies (VMC and DMC) with the same expansion length if the CSFs are chosen in the presence of a Jastrow correlation function. Since we dispensed with the optimization of the molecular orbitals, these findings show that the selection is altered when a Jastrow factor is present, justifying the need for a Jas-sCI scheme.

Lastly, we also included the optimization of the MOs in the Jas-sCI scheme in order to evaluate the impact of improved orbitals on the selection, see Table 7. Again, for the Jas-sCI entries, we started with larger expansions (initial cutoff) at different optimization levels (initial opt level), truncated the wave functions (after the optimization) to 168 CSFs, and subsequently reoptimized the Jastrow, MO, and CI parameters. Let us start by discussing the results for the smaller initial wave function (0.0050). First, the fully optimized initial wave function provides lower VMC and DMC energies than the CIPSI selection. For the Jas-6+CI initial wave function, we do

not see a significant improvement (for VMC and DMC). More importantly, though, is the finding that when the CSFs are selected using improved orbitals, both lower VMC and DMC energies—compared to the Jas-6+CI initial optimization level—are obtained. This shows that the orbitals influence the selection of the CSFs. We want to make one last comparison for this smaller initial wave function. The DMC energy for the fully optimized CIPSI-Jastrow wave function, truncated at a threshold of 0.0050 (266 CSFs) is equal to −75.9158(5) E_h . Comparing this energy to the ones from Table 7 (third and fourth row), we see first that, for the (0.0050, Jas-6+CI) initial wave function, the same—within the statistical error—DMC energy can be obtained with a more compact wave function (168 vs 266 CSFs). Second, and more importantly, a lower DMC energy is attained (for a smaller expansion size) if one starts with a fully optimized wave function (0.0050, Jas-6+MO+CI). Note, however, that the DMC energy of −75.9158(5) E_h is not significantly better than the CIPSI-Jas (168 CSFs, Jas-6+MO+CI) DMC energy of −75.9151(5) E_h although it contains more CSFs.

For the larger initial expansion (0.0025) at the Jas-6+CI initial opt level, we obtain a Jas-sCI DMC energy of −75.9161(5) that is lower than the DMC CIPSI-Jas energy (−75.9151(5)), but that is similar to the one of the initially fully optimized Jas-sCI wave function (0.0025, Jas-6+MO+CI) (−75.9162(5)). This result shows that it might not be necessary to make the CSF selection with improved orbitals, since the MO optimization after the final Jas-sCI iteration might be able to compensate for the effect of improved initial orbitals. This implies that it might be possible to design an iterative Jas-sCI algorithm without needing to (costly) optimize the molecular orbitals at each iteration for the selection of the CSFs.

Having observed an improvement in the energy for some Jas-sCI approaches, we decided to investigate the respective wave functions in more detail, trying to gain more insight into how the selections are different from one another. The CIPSI and the (0.0050, Jas-6+MO+CI) Jas-sCI selections differ by 36 of the total 168 CSFs. When analyzing the occupation of these CSFs, no systematic preference can be deduced. For the Jas-sCI selection, configurations that include excitations e.g. out of orbital $2a_g$ are removed while CSFs with excitations into orbitals $1b_{3g}$ and $1b_{2g}$ (primarily from $1b_{2u}$ and $1b_{3u}$) are introduced into the wave function. The orbitals can be found in the Supporting Information.

Tailoring CIPSI expansions has been successfully done in various studies,^{18,22–26} corroborating the promising nature of the CIPSI-Jastrow approach. In these studies, it was yet never tried to go beyond the CIPSI selection. Our findings emphasize, however, the importance of choosing the expansion in the presence of the Jastrow factor, which is why, in a next step, we aim at expanding (small) VMC optimized wave

Table 6. VMC and DMC Energies in E_h for Different Wave Function Ansätze^a

	initial cutoff (no. CSFs)	initial opt level	$E_{\text{VMC}}(\text{Jas-6+CI})$	$E_{\text{DMC}}(\text{Jas-6+CI})$
CIPSI-Jas/HF			−75.8820(2)	−75.9121(5)
Jas-sCI/HF	0.0050 (266)	Jas-6+CI	−75.8842(2)	−75.9127(5)
	0.0025 (528)	Jas-6+CI	−75.8849(3)	−75.9145(5)

^aThe expansions were truncated at 168 CSFs. For the CIPSI-Jas wave function, the Jastrow and CI parameters were optimized. For the Jas-sCI entries, the CIPSI wave functions for different thresholds (see initial cutoff) were truncated (to 168 CSFs) and the Jastrow and CI parameters were re-optimized. The initial orbitals—used for the CIPSI calculation—are Hartree–Fock orbitals.

Table 7. VMC and DMC Energies in E_h for Different Wave Function Ansätze^a

	initial cutoff (no. CSFs)	initial opt level	$E_{\text{VMC}}(\text{Jas-6+MO+CI})$	$E_{\text{DMC}}(\text{Jas-6+MO+CI})$
CIPSI-Jas/HF			−75.937(3)	−75.9151(5)
Jas-sCI/HF	0.0050 (266)	Jas-6+CI	−75.8941(3)	−75.9154(5)
		Jas-6+MO+CI	−75.8951(3)	−75.9165(5)
Jas-sCI/HF	0.0025 (528)	Jas-6+CI	−75.8941(3)	−75.9161(5)
		Jas-6+MO+CI	−75.8946(3)	−75.9162(5)

^aThe expansions were truncated at 168 CSFs. For the CIPSI-Jas wave function, the Jastrow, MO, and CI parameters were optimized. For the Jas-sCI entries, the CIPSI wave functions for different thresholds (see initial cutoff) as well as different optimization levels (see initial opt level) were truncated (to 168 CSFs) and fully reoptimized. The initial orbitals—used for the CIPSI calculation—are Hartree–Fock orbitals.

functions by generating new CSFs rather than reducing the size of existing expansions.

CONCLUSION

In this work, we studied truncated CIPSI-Jastrow wave functions for the $^1\Sigma_g^+$ ground state of C_2 with variational and diffusion quantum Monte Carlo. The truncation was carried out based on the absolute value of the CI coefficients. The Jastrow parameters, molecular orbitals, and CI coefficients were either partially or fully optimized with respect to the variational energy. A considerable decrease in, both, the VMC and DMC energies was observed by appending a Jastrow correlation function to the truncated CIPSI expansions, legitimizing the use of a Jastrow factor to treat the short-range dynamic electron correlation. CSF-based wave functions are favored over determinant-based ones because first, the reduced number of CI parameters leads to less costly optimizations and second, lower energies are obtained. The truncated CIPSI-Jastrow-DMC approach provides—for the smaller coefficient thresholds—very accurate absolute and bond dissociation energies compared with different ansätze from literature. We demonstrated that, for the dissociation energy, the size inconsistency, arising from the rather aggressive cutoff values, can be resolved by choosing the same wave function norms as thresholds for C_2 and the carbon atom. Furthermore, the CI picture is drastically changed by employing a Jastrow factor and reoptimizing the antisymmetric part of the wave function. The comparison of Hartree–Fock, Kohn–Sham, and CASSCF orbitals shows that for a given threshold, mostly lower VMC (and DMC) energies are obtained with smaller expansions if the initial orbitals result from a correlated calculation. The key result of this work is, however, that lower VMC and DMC energies—compared to the CIPSI selection—can be obtained with the same expansion size, if the CSFs are selected in the presence of a Jastrow correlation function, paving the way toward a Jastrow selected CI scheme within the QMC framework.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.2c01229>.

VMC and DMC energies of the truncated CIPSI (no Jas) wave functions for C_2 , VMC and DMC energies of the truncated CIPSI-Jastrow wave functions of C_2 for each ansatz (Det, CSF, Jas-4, Jas-6), carbon atom VMC and DMC energies for both truncation schemes for the dissociation energies, comparison of the DMC energies for the determinant and CSF expansions, relevant HF and VMC optimized orbitals of C_2 , DMC extrapolation with respect to the norm of the wave function, VMC and

DMC energies of C_2 using different sets of initial orbitals (B3LYP, PBE0, CAS(8,8)) in the CIPSI calculations, several figures relating to the discussion of the initial orbital choice (ZIP)

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Notes

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